Healthcare & Data Science: Malaria Detection with CNN's

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Content

Identifying the Problem

Globally, an estimated 282 million cases of malaria were reported in late 2024, with 610,000 deaths. Roughly 9 million more cases than the previous year.
(World Health Organization 2025)

Untreated Malaria can be fatal very quickly, sometimes within 24 to 48 hours of symptoms starting.
(World Health Organization 2025)

Methods used to detect malaria could take up to 15 minutes, another could take days to weeks in order to confirm the species of malaria.
(CDC 2024)

Our Opportunity

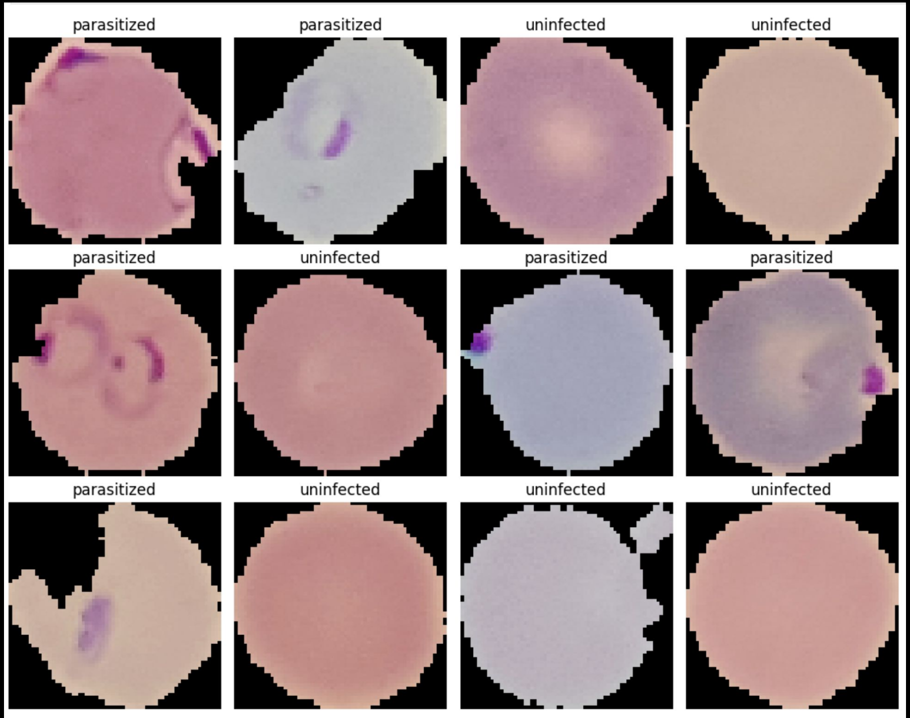
Can we provide healthcare providers with a tool that accurately detects whether a patient is infected with malaria or not in a timely manner?

Key Findings & Insights

Within the 27,554 microscopic images of red blood cells that were provided, parasitized cells appeared to contain a **ring-like structure** within.

Given the large, **unlabeled** dataset, the 13,882 images that contain the ring-like structure is the **key distinguisher** for our model to learn.

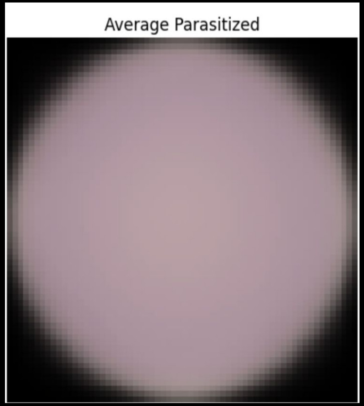
Due to the various sizes of microscopic images provided, **data preprocessing** is a must for fast training and improved accuracy.

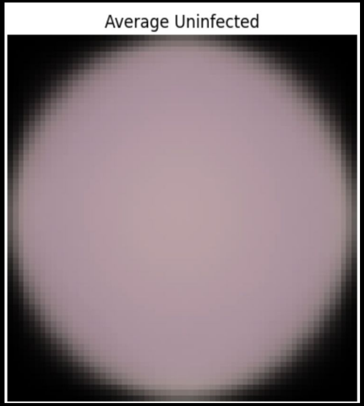


Key Findings & Insights
cont'd

Post-data preprocessing, averages of both parasitic and non-parasitic microscopic images result in a single image with a **flat, uniform** color.

Identical coloring can mean pixel values are within a **reasonable range** and color across images at each pixel position is **consistent**, which means the data preprocessing was **successful.**



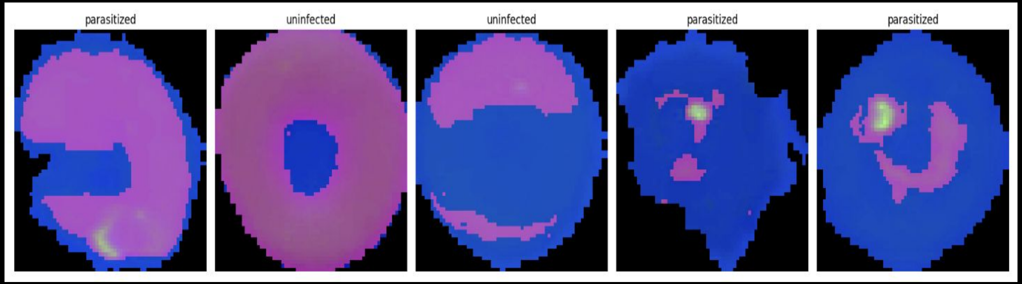


Key Findings & Insights

cont'd, final

Converting the microscopic images from RGB to HSV, highlights the ring-like structures in parasitic cells.

Highlighting the ring-like structure could **improve** the model's ability to learn at a faster rate.



Sequential Convolutional Neural Network (CNN)

1

Efficient Feature Learning
Apply a set of learnable filters to detect features.

2

Maintain Robust Performance & Invariance.
Recognize features regardless of minor shifts.

3

Generate Accurate Predictions
Determine final image representation using learned features for accurate prediction.

Assessing the Performance & Accuracy

97.39%

Average accuracy between training and validation sets.

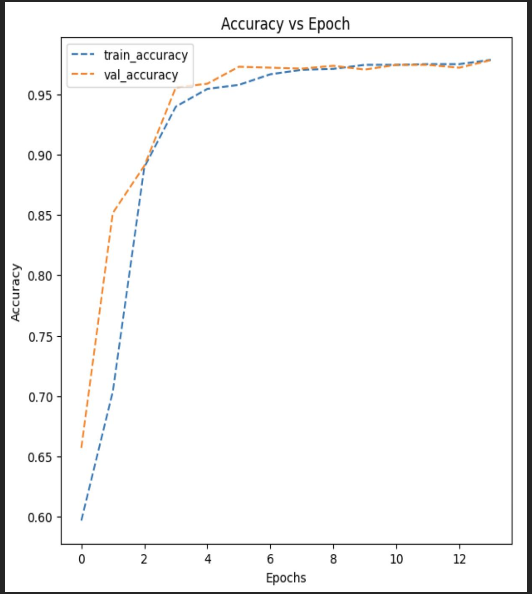
95-98%

Accuracy required in high stake, direct patient care scenarios.

Overall Assessment

Achieved strong performance quickly. Accuracy increases, as data is trained, or rather, passed through the algorithm.

Training and Validation Accuracy:



The training and validation accuracy curves showed good convergence with minimal overfitting initially.

Strategic Advantages

Accelerated Clinical Outcomes

Diagnose patients much quicker than traditional manual methods,

Operational Efficiency & Staff Optimization

Reduce the workload on laboratory technicians and medical professionals.

Market Expansion & Health Equity

Accessible in remote or under-resourced areas where specialized expertise may be limited.

Managing Risks

Regulatory Compliance

Obtaining necessary approvals from health authorities.

Technical Integration with Existing Systems

Integrate the new model with diverse existing hospital information systems and microscopy equipment.

Building User Acceptance and Trust

Healthcare professionals are hesitant to fully trust AI diagnostics .

Execution

1

Stakeholder Alignment

Consult with healthcare providers.

2

Controlled Case Study

Launch a clinical case study to gain trust.

3

Validation & Safety

Assess the model's impact.

4

Implementation

Invest in modern infrastructure.

Thank you

Appendix

Artifacts

```
#create sequential model
model = Sequential()

model.add(Conv2D(filters = 32, kernel_size = 2, padding = "same", activation = "relu", input_shape = (64, 64, 3)))

model.add(MaxPooling2D(pool_size = 2))

model.add(Dropout(0.2))

model.add(Conv2D(filters = 32, kernel_size = 2, padding = "same", activation = "relu"))

model.add(MaxPooling2D(pool_size = 2))

model.add(Dropout(0.2))

model.add(Conv2D(filters = 32, kernel_size = 2, padding = "same", activation = "relu"))

model.add(MaxPooling2D(pool_size = 2))

model.add(Dropout(0.2))

model.add(Flatten())

model.add(Dense(512, activation = "relu"))

model.add(Dropout(0.4))

model.add(Dense(2, activation = "softmax")) # 2 represents output layer neurons

model.summary()
```

Artifacts

cont'd

```
Epoch 1/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 30s 170ms/step - accuracy: 0.5659 - loss: 0.6838 - val_accuracy: 0.6570 - val_loss: 0.6382
Epoch 2/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 38s 152ms/step - accuracy: 0.6683 - loss: 0.6215 - val_accuracy: 0.8516 - val_loss: 0.4928
Epoch 3/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 40s 146ms/step - accuracy: 0.8532 - loss: 0.3950 - val_accuracy: 0.8906 - val_loss: 0.2079
Epoch 4/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 25s 155ms/step - accuracy: 0.9339 - loss: 0.1604 - val_accuracy: 0.9555 - val_loss: 0.1407
Epoch 5/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 25s 155ms/step - accuracy: 0.9581 - loss: 0.1184 - val_accuracy: 0.9586 - val_loss: 0.1408
Epoch 6/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 41s 156ms/step - accuracy: 0.9651 - loss: 0.1116 - val_accuracy: 0.9727 - val_loss: 0.1208
Epoch 7/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 23s 146ms/step - accuracy: 0.9714 - loss: 0.0937 - val_accuracy: 0.9719 - val_loss: 0.0979
Epoch 8/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 42s 152ms/step - accuracy: 0.9721 - loss: 0.0901 - val_accuracy: 0.9711 - val_loss: 0.0861
Epoch 9/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 43s 162ms/step - accuracy: 0.9739 - loss: 0.0801 - val_accuracy: 0.9734 - val_loss: 0.0897
Epoch 10/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 40s 153ms/step - accuracy: 0.9749 - loss: 0.0767 - val_accuracy: 0.9703 - val_loss: 0.0850
Epoch 11/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 41s 151ms/step - accuracy: 0.9759 - loss: 0.0686 - val_accuracy: 0.9742 - val_loss: 0.0848
Epoch 12/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 41s 153ms/step - accuracy: 0.9774 - loss: 0.0669 - val_accuracy: 0.9742 - val_loss: 0.0757
Epoch 13/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 23s 145ms/step - accuracy: 0.9792 - loss: 0.0587 - val_accuracy: 0.9719 - val_loss: 0.0806
Epoch 14/20
160/160 ━━━━━━━━━━━━━━━━━━━━ 41s 143ms/step - accuracy: 0.9803 - loss: 0.0556 - val_accuracy: 0.9781 - val_loss: 0.0799
```

Artifacts

Cont'd final

	precision	recall	f1-score	support
0	0.98	0.97	0.98	3752
1	0.96	0.98	0.97	2600
accuracy			0.97	6352
macro avg	0.97	0.97	0.97	6352
weighted avg	0.97	0.97	0.97	6352